

A Sliding Window Based Approach With Majority Voting for Online Human Action Recognition using Spatial Temporal Graph Convolutional Neural Networks

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ABSTRACT

Nowadays, Human Action Recognition (HAR) has become an important issue since it is widely used in video surveillance, human-robot collaboration in industry, etc. Developing such accurate and efficient algorithms remains a difficult task because of the high variability of the human shapes, postures, as well as the complexity of their movements but more importantly when using continuous/untrimmed data streams. Since HAR from Segmented/trimmed sequences has been intensively studied and developed in the recent years, Online HAR in the other hand remains a challenging task and is less developed. In this paper, we propose a Sliding Window and Majority Voting skeleton-based approach for Online HAR using Spatial Temporal Graph Convolutional Neural Networks (STGCN-SWMV). Our method is evaluated on two Online skeleton-based datasets named OAD and UOW. In comparison with existing methods, the obtained results exceed state-of-the-art algorithms and show the efficiency of the proposed approach.

KEYWORDS

Online Human Action Recognition, Spatial Temporal Graph Convolutional Neural Networks, Sliding Window, Majority Voting

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1 INTRODUCTION

Online Human Action Recognition (HAR) has attracted many researchers in recent years due to its wide applications in video surveillance, human computer interaction, robotics, and production industries [6, 31] as it can be used to make human and robot collaborate, improve workplace safety and make production systems more flexible and thus increasing products quality. The developed approaches can be grouped into four main categories according to the data modality used: RGB based methods [30], Depth Maps-based methods [31], Skeleton-based methods [34], and Hybrid methods where at least two modalities are combined [19].

Even if RGB-based approaches are the most basic methods used for action recognition, there are sensitive to high variability of human shapes or postures, backgrounds clutter and camera position. Moreover, if used at work place, it is an intrusive tool.

RGB-based approaches are the most basic methods used for action recognition. Conventional cameras have represented the acquisition systems on which most methods have been based. Such methods are quiet efficient, if enormous amount of data is available. Actions can be easily be predicted from RGB data and information such as postures or poses can be easily deducted from images or videos [30]. However, high variability of human shapes or postures and backgrounds clutter in such data can be an issue. Moreover, cameras can be sensitive to illumination changes, occlusions and orientation changes. Finally, camera is intrusive at work place.

Methods based on Depth-Maps rely on the extraction of representative hand-crafted feature descriptors from the sets of points in the depth image/video. Such features can be shallow and also dataset-dependent. Thus, modeling the dynamics of an action is an important challenge which can be solved differently depending on the approach. Skeleton features can be extracted from depth-maps and be devised to capture both spatial and temporal configurations but such features are hand-crafted, thus resulting in limited expressive power and difficulties of generalization [31].

2D/3D Skeleton-based approaches have become more popular in recent years due the advantages they have over conventional RGB data. These approaches are increasingly used since they are effective, reliable and convey more representative information considering the fact that they overcome cluttered backgrounds and are insensitive to illumination changes unlike RGB-based methods [31].

Skeleton-based methods are reliable to estimate the body silhouette more easily and do not need huge amount of data to perform the recognition task [34].

Hybrid solutions combine information from at least two data modalities (Skeleton, RGB, or Depth-Maps) to model actions which can improve the performance of the action recognition [19].

Tools like OpenPose [5] or AlphaPose [33] can be used to extract skeleton data from RGB streams in real time which will allow fast and accurate pose estimations. Skeleton-based approaches are more efficient against background clutter problems than 2D/3D images or videos. Also, skeletal data provide more accurate representation of the human body posture. In addition, skeletal data are more suitable for online action recognition since they have low dimensions, which reduces both learning and testing times. This is why, in the following, we are interested in skeleton-based methods only.

Following this categorization, existing methods for action recognition using skeletal data are subsequently analyzed.

A human action is naturally characterized by the evolution of the human body over time. Skeletal data, containing 3D positions of the different body joints, provide an accurate representation of the posture of the human body and can provide more comprehensive information for action recognition compared to 2D images/videos. These characteristics can be easily extracted and tracked from skeletal data which provide local information about the human body. However, even if precise 3D positions of different joints are available, the action recognition task remains difficult because of the high temporal and spatial variations the way an action is performed.

These considerations motivate us to approach the online human action recognition problem due to its wide applications in video surveillance, human computer interaction, robotics and production industries [31], as it can be used to make human and robot collaborate, improve workplace safety and make production systems more flexible and thus increasing products quality [6]. Thereby, in this paper we propose a sliding window-based approach for online human action recognition using Spatial Temporal Graph Convolutional Neural Networks (ST-GCNs). To do so, we extend classic Graph Convolutional Networks (GCNs) to the space-time domain in order to address the online action recognition problem. Subsequently, we added the experimental analysis to evaluate the effectiveness of the proposed method tested on two online skeleton-based datasets named OAD [15] and UOW [28].

The main contributions of this paper are summarized as follows:

- We investigate the Online HAR problem for continuous streaming skeletal data which is challenging and very less developed in the literature than Segmented HAR by leveraging a Sliding Window and Majority Voting based approach using a ST-GCN.
- Experimental results demonstrate the effectiveness of our method evaluated on two Online skeleton-based datasets named OAD [15] and UOW [28] exceeding state-of-the-art performances in Accuracy and Intersection over Union (IoU) on Online HAR.

The code and the pre-trained models of the STGCN-SWMV method are made publicly available¹.

¹<https://github.com/mejdidalallel/STGCN-SWMV>

The rest of this paper is organized as follows. In Section 2, we review the related work on skeleton-based action recognition and formulate the segmented/online action recognition problems. In Section 3, we illustrate the details of our proposed method. Experimental results and discussions are presented in Section 4. Finally, concluding remarks and future works are given in Section 5.

2 RELATED WORK

Human action recognition have drew researchers attention in the last few years and it can be divided into two types: segmented and online recognition [6]. By segmented (or trimmed) recognition, which is the most common type of recognition used by most of researchers, we refers to the fact of, given an action sequence, determining if a frame or a set of frames, at a specific time, matches an action among the preset actions classes, therefore the recognition step is performed after observing the full action sequence [23]. Such methods are standard and are rather easy to implement as they do not need deep complex networks to perform the recognition task. For instance, in [36], another algorithm is needed to segment the data streams, this is the major limitation of this approach because segmenting data streams makes the algorithm heavy which makes the approach not suitable for real-time scenarios. On the other hand, online recognition, consists of predicting an action on the fly from-within continuous/untrimmed data streams without using any further information i.e alongside with predicting the action label, online recognition also aims to predict the start and the end points of each action within long series or sequences of different/similar actions. Online recognition is a lot more complex task than segmented recognition since it aims to detect the action after partially observing its sequence and without using any further information. Also, besides of determining an action label, it should also predict the start and end time of each action in a long series of actions with lots of background and noises which makes such methods a lot more complex [15]. Segmented action recognition are limited to the fact that they need to wait for the completion of the action sequence in order to recognize it and this is not desirable for real-time applications which needs to predict the action on the fly without using any further information. This is why in our method we proposed an online action recognition algorithm.

In [15], authors have used an end-to-end Joint Classification-Regression Recurrent Neural Network (JCR-RNN) for Online action detection and forecasting. They performed frame-wise action classification to detect actions. Then they localized the start and end points of each action using a regression sub-network. Specifically, a Long Short-Term Memory (LSTM) sub-network was used as recurrent layers for feature learning and long-range temporal dynamics modeling based on the baseline method Deep LSTM in [38]. The main limitation of using 2D skeletal data is that it makes some actions look ambiguous when dropping the depth dimension especially when subjects interact with objects. Based on [15], same authors developed a new method integrating both RGB and Skeleton networks for Online HAR [19]. They expanded temporal modeling networks to capture specific characteristics from different data modalities.

In [2], authors used RGB-D based Convolutional Neural Network (CNN) features as spatial-temporal contexts in order to reduce

potential noise in skeleton estimation using Random Forests (RF) for Online Action Recognition.

In [34], authors used a Spatial Temporal Graph Convolutional Neural Network (ST-GCN) for segmented action recognition. They have proposed a new representation of skeletal sequences for action recognition by extending Graph Convolutional Networks (GCNs) to the spatial-temporal domain. In fact, GCNs generalize the operation of convolution from traditional data like images or videos into graph data. This operation consists in teaching a function to generate the representation of a node by aggregating its own characteristics and the characteristics of its neighbors. During a given action, the joints move in local groups, which they call "body parts", which facilitates their modeling. This is due to the fact that some parts limit the modeling of joints trajectories within "local regions" with respect to the complete skeleton, thus forming a hierarchical representation of skeletal sequences. Given such sequences as 2D or 3D coordinates, a spatio-temporal graph is then constructed where joints are represented by the graph nodes, and the natural connectivity in the human body structures are represented by the graph edges. The construction is done in two steps. First, joints in one frame are connected by edges according to the connectivity of the skeletal structure of the human body. Then, each joint will be connected to the same joint in the next frame. ST-GCNs perform well in segmented scenarios but are not adapted to online ones.

To help clarify the differences, we will first discuss segmented recognition.

Segmented recognition: aims to determine if a given segmented video containing N frames $V_{seg} = \{f_1, \dots, f_N\}$, matches an action among the K preset action classes. Therefore, the action recognition step is carried out after observing the full action sequence [6]. Thus, determining an action label at each time slot means maximizing its posterior probability which can be formulated as:

$$y^* = \underset{y}{\operatorname{argmax}} P(y|V_{seg}) \quad (1)$$

where y represents the possible output action label vector of a segmented video V_{seg} . Therefore, the action label with the highest probability $P(y|V_{seg})$ is selected as the actual label of the video sequence V_{seg} .

Online recognition: in opposition to segmented action recognition, which uses the entire video sequence to determine the action label, online recognition aims to recognize an action within a long sequence of different/similar actions on the fly at the earliest i.e without using any additional information [6]. Online recognition can be formulated as:

$$y_t^* = \underset{y_t}{\operatorname{argmax}} P(y_t|f_1, \dots, f_t) \quad (2)$$

Online recognition has two main challenges compared to segmented recognition. Apart from the fact of having a higher computational cost, it also needs to recognize the action in progress, while being able to segment it, which means detecting the beginning and the end of each action.

Both segmented and online recognition are illustrated in Figure 1.

In [27], authors developed a framework for online real-time multiple spatio-temporal action localisation and classification. They

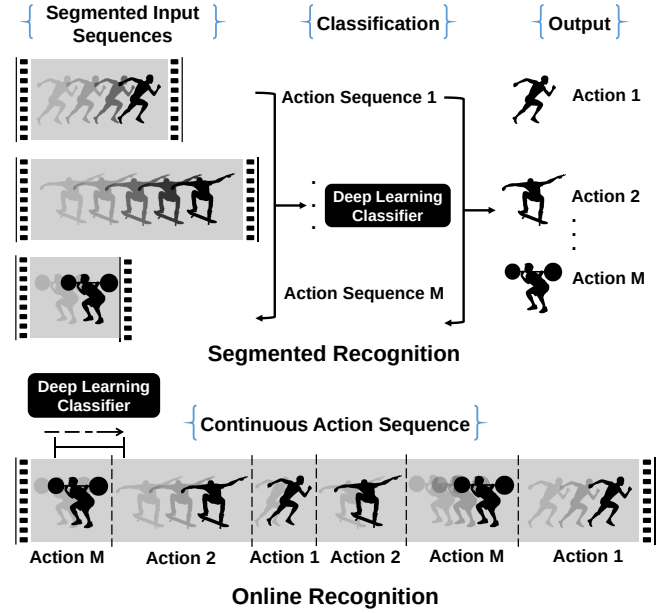


Figure 1: Segmented/Online Action Recognition

have used a real-time Single Shot MultiBox Detector (SSD) Convolutional Neural Networks (CNNs) for regression and classification of the detected boxes in each frame of the video sequences that may contain a relevant action. SSDs are used to approximate the output of the detected bounding boxes to a set of boxes over different aspect ratios and scales by producing adjustments to the boxes so they match the object/human shape. As for online recognition, they developed an online algorithm to construct and label the actions tubes from the SSDs frame-wise detections.

In [17], authors developed a SlowFast-GCN framework by combining the advantages of both ST-GCNs and SlowFastNets with dynamic human skeleton. They have used two streams, the first stream is used to capture the motion changes in human actions/movements and the second one is used to capture static semantics. Human skeleton features from two different time dimensions are then merged. Training using such features seems to improve the accuracy of the human action recognition.

Even though online action recognition is crucial for many nowadays applications, few works have been proposed addressing this matter. Therefore, in this paper, we investigate the problem of online action recognition for continuous skeletal data streams by leveraging a Sliding Window and Majority Voting based approach using Spatial Temporal Graph Convolutional Networks (STGCN-SWMV).

3 ST-GCN WITH A SLIDING WINDOW AND MAJORITY VOTING APPROACH (STGCN-SWMV)

Based on the work of [34], we propose a Spatial Temporal Graph Convolutional Neural Network using a Sliding Window and Majority Voting approach to address the online human action recognition problem.

In the following, we first briefly review GCNs and ST-GCNs to make the paper self-contained. Then we introduce our proposed Spatial Temporal Graph Convolutional Neural Network with a Sliding Window and a Majority Voting approach (STGCN-SWMV) for online action recognition.

3.1 Overview of GCNs and ST-GCNs

Neural Networks (NNs) have been hugely successful over the past decade [1]. However, the first variants of neural networks should only be implemented using Regular or Euclidean data, whereas much real-world data has underlying graph structures that are not Euclidean. The non-regularity of data structures has led to recent advances in Graph Neural Networks (GNs). In recent years, different variants of Graph Networks (GNs) have been developed, Graph Convolutional Networks (GCNs) being one of them. GCNs are also described as one of the basic variants of Graph Networks (GNs) [32].

GCNs perform similar operations as Convolutional Neural Networks (CNNs), where the model learns features by inspecting neighboring nodes. The main difference between CNNs and GCNs is that CNNs are specially designed to work on regular Euclidean structured data such as images, while GCNs are the generalized version of CNNs where the number of node connections varies and the nodes are not ordered (Irregular on Non-Euclidean structured data) [4].

More formally, a GCN is a neural network that operates on graphs. A graph G is represented by a set V of N nodes $n_i \in V$ and a set of edges E where $e_{i,j} = (n_i, n_j)$. Given a graph $G = (V, E)$, a GCN takes as input a $N \times F^0$ feature matrix (where F^0 is the number of input features for each node) and a $N \times N$ matrix representation of the graph structure generally in the form of an adjacency matrix A . As for the output, it generates a $N \times F$ feature matrix (where F represents the number of output features per node).

More generally, a hidden layer in the GCN can be written as $h^i = f(h^{i-1}, A)$ where h^0 is the input feature matrix and f represents the propagation function. Each layer h^i is a $N \times F^i$ feature matrix where each row is a feature representation of a node. Features are aggregated at each layer to form the next layer features using the propagation function f . At each consecutive layer, features become more and more abstract.

GCNs themselves can be classified into two main groups; Spectral Graph Convolutional Networks, in which graph convolution locality is considered in the shape of spectral analysis [11, 16] and Spatial/Temporal Graph Convolutional Networks, where the operation of convolution is applied on the graph nodes and their neighbors [12, 24]. As described in Figure 2, the operation consists in teaching a mapping function f to generate the representation of a node n_i by aggregating its proper features x_i and the features of its neighbors x_j over time and space [32]. Based on the work of [34], other improved methods have been proposed as in [14, 25, 37].

In this article, we will focus on Spatial Temporal Graph Convolutional Networks.

3.2 Sliding Window Approach

Sliding windowing is the most widely used technique in the literature for Online HAR since it does not require data pre-processing

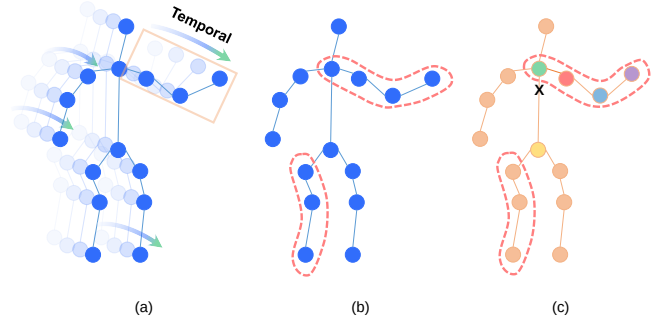


Figure 2: (a) Illustration of the Spatial-Temporal Graph of a skeleton sequence denoting human movement in space and time used in ST-GCNs. Blue dots denote the skeleton joints and arrows represent their connection in consecutive frames. (b) Sampling strategy in a convolution layer for a single frame. (c) Illustration of the mapping strategy.

and it is easy to use in real-time [7]. This technique consists of splitting the signal to be analyzed into a series of fixed-size windows. An overlay between generated series can be used, it can vary from 10% to 90%. This windowing is particularly used for periodic (e.g. walking or running) and static (e.g. resting) HAR [3, 7, 29].

The size of the Sliding Window (SW) has a major impact on the recognition accuracy. In this paper, the choice of the SW size as well as the window-overlapping rate is done in order to guarantee satisfactory characterization of the transitional activities. Therefore, the SW size for each dataset we tested on is set to the mean duration of all the performed actions which gives the best results among other tested SW sizes. Choosing a SW size equal to the mean duration of all actions is very common in the literature [3, 22, 29]. The SW size, W , can be formulated as:

$$W = \frac{1}{N} \sum_{i=1}^N Seq_{duration}(i) \quad (3)$$

where N denotes the number of the input action sequences and $Seq_{duration}$ denotes the duration of the i^{th} action sequence.

The step size of the SW, SW_{step} , represents the length of the sequence that moves between each window. For both datasets, we set SW_{step} to 1 frame resulting in an overlap between the SWs. This way, we assure that all actions including ephemeral ones are well captioned.

The structure of the proposed STGCN-SWMV module is shown in Figure 3.

3.3 STGCN-SWMV for Online Recognition Task

A new skeletal sequence representation for action recognition was proposed in [34] by extending graph convolutional networks to the spatial temporal domain. During a given action, the skeleton joints move in local groups called body parts which makes their modeling easier. This is due to the fact that the parts limit joints trajectories modeling within local regions in comparison to the entire skeleton

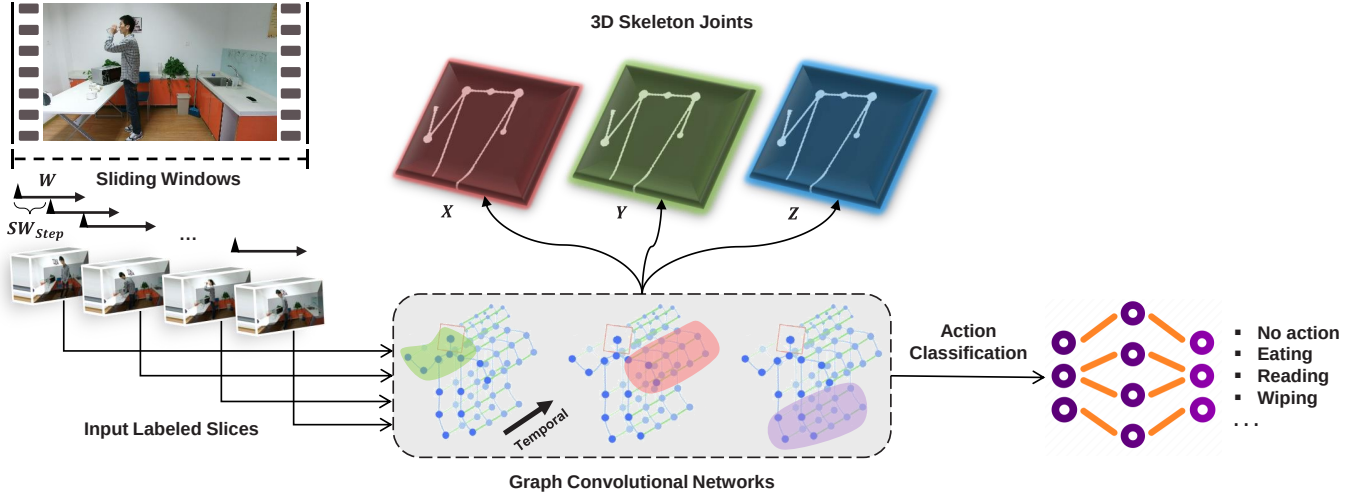


Figure 3: The structure of the proposed Spatial Temporal Graph Convolutional Network for Online Action Recognition

and thus develop a hierarchical representation of the skeleton sequences. The spatial edges conforms to the natural connectivity of joints in the body and the temporal edges connects the same joints across consecutive frames (See Figure 2). Thus, the trajectory of a particular joint over time is characterized by all the edges of the graph. Multiple layers of spatial temporal graph convolutions are constructed thereon generating higher level feature maps, which allow information to be integrated along both spatial and temporal dimensions. Inputs of the STGCN-SWMV module are the coordinate vectors of the joints on the graph nodes. Given sequences of the skeleton joints, the STGCN-SWMV module automatically captures the patterns embedded in the spatial configuration of the skeleton joints and exploits the correlation between each joint by constructing a spatial temporal graph as well as their temporal dynamics featuring both intra-body and inter-frame connection.

Following their work, we coupled ST-GCNs with a Sliding Window and Majority Voting approach to address the online action recognition problem.

During a given action, the skeleton joints move in local groups called body parts which makes their modeling easier. This is due to the fact that the parts limit joints trajectories modeling within local regions in comparison to the entire skeleton and thus develop a hierarchical representation of the skeleton sequences.

The input of the ST-GCN module are the coordinate vectors of the joints on the graph nodes. Given sequences of the skeleton joints, the ST-GCN module exploits the correlation between each skeletal joint by constructing a spatial temporal graph. The equation from this ST-GCN [34] for the i^{th} node at a time t , n_{ti} , is formulated as follows [26]:

$$f_{out}(n_{ti}) = \sum_{n_{tj} \in B(n_{ti})} \frac{1}{Z_{ti}(n_{tj})} f_{in}(n_{tj}) \cdot w(l_{ti}(n_{tj})) \quad (4)$$

where $B(n_{ti})$ represents the neighbors of the node n_{ti} , Z represents the cardinality of the corresponding subset, f_{in} is the input feature map and w is the weight function similar to the kernel of

a 2D convolution. For the weight function, in 2D convolution, a fixed grid naturally exists around the central location. The neighbor pixels can have a fixed spatial order. In the ST-GCN, it simplifies the process by configuring the spatial partitioning.

3.4 Majority Voting (MV)

The general weighted Majority Voting can be written as:

$$\hat{y} = \underset{i \in K}{\operatorname{argmax}} \sum_{j=1}^{N_v} \alpha_j C \quad (5)$$

where $\hat{y}$ is the predicted class label at frame t , K is the number of classes, N_v is the number of predictions or votes, α_j is a weighting coefficient applied to the classifier and C is the classifier one-encoded vector output of the K class labels. The number of predictions or votes, N_v , for one specific frame is equal to K .

In the proposed method, we are using this Majority Voting approach. First, the same classifier C is giving a prediction several times for the same frame f_t . Indeed, the predicted output label $\hat{y}_t$ of the classifier depends on the placement of the input data $\mathbf{x}_t$ inside the SW as described in Figure 4.

Therefore, we can consider the SWs as N_v different classifiers. The SW classifier can be expressed as:

$$\begin{aligned} C(t, W) &= \hat{\mathbf{y}}_{t, W} \\ &= [\hat{y}_t, \hat{y}_{t+1}, \dots, \hat{y}_{t+i}, \dots, \hat{y}_{t+W-1}] \\ &= C([\mathbf{x}_t, \mathbf{x}_{t+1}, \dots, \mathbf{x}_{t+i}, \dots, \mathbf{x}_{t+W-1}]) \end{aligned} \quad (6)$$

where $\hat{\mathbf{y}}_{t, W}$ is a vector of the predicted classes for each input data from frame t with a SW of size W . When using ST-GCN as a classifier, the outputs are the same for each frame. As an example, in Figure 4-1, the output action label of the frame $f_{t=5}$ is not the same when using data from frame f_{t-w+1} to f_t than when using data from frame f_{t-w+2} to f_{t+1} . Therefore, $\hat{o}_t$ accumulates the voted class given by each classifier from $C(t - W + 1, W)$ to $C(t, W)$ (See Figure 4-1 & 2)

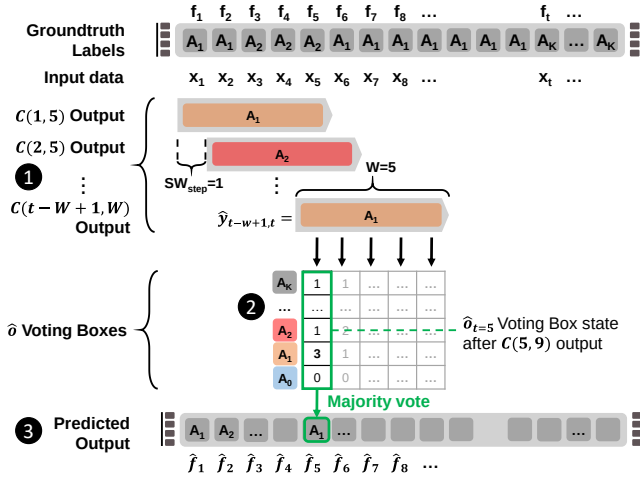


Figure 4: Illustration of the Sliding Window approach with the Majority Voting principle

Finally, the actual predicted action class of a frame $\hat{f}_t$ is set to the action that received the highest number of vote (see Figure 4-3). The results are extracted with this equation:

$$\hat{y}_t = \text{mode}\{\hat{o}_t\} \quad (7)$$

Finally, this Majority Voting proposition allows smoothing the final output labels because actions are voted from different point of views of the timeline.

4 EXPERIMENTS

In this section, we evaluate the Online HAR performance of the proposed method on two challenging Online skeleton-based datasets.

- The Online Action Detection Dataset (OAD) [15], which was captured using the Microsoft Kinect V2 sensor (RGB-D and skeleton joints modalities), includes 59 long sequences where each subject repeats arbitrary 10 different actions.
- The Online Action3D Dataset (UOW) [28] consists of 20 action sequences of skeleton videos recorded with a Microsoft Kinect V2 where 20 participants were invited to perform the different actions in a random order and each action is repeated 3 to 5 times.

Other HAR datasets such as THUMOS [10] or ActivityNet [9] were not considered in the evaluation of our method since action classes in these datasets were either long activities, comprising humans in a close-up view or does not include humans at all. Therefore, extracting human skeletons from such data was not possible to evaluate the proposed approach.

4.1 Parameter Settings

For the OAD dataset [15], we set the SW size to 40 frames with an original fps equal to 8. As for the UOW dataset [28], we set the SW size to 50 frames with an original fps equal to 20. For both datasets, we used the same train, validation and test sequences set in the original datasets papers.

4.2 Action Recognition Evaluation Metrics

To evaluate the performance of our Online HAR method, we used three different evaluation metrics besides Accuracy.

- Intersection-over-Union (IoU): is used to determine if a predicted interval for an action is correct or not. The predicted interval is considered as correct when an overlapping ratio, IoU , between the prediction I and the groundtruth I^* label exceeds a defined threshold θ . IoU can be expressed as follows:

$$IoU = \frac{|I \cap I^*|}{|I \cup I^*|} \quad (8)$$

where $I \cap I^*$ denotes the intersection between prediction and groundtruth labels and $I \cup I^*$ denotes their union. In all of our tests, we set the overlapping ratio threshold θ to 0.8.

- F1-Score: is used to interpret the weighted average of the precision and recall. It reaches its best value at 1 and worst score at 0. The advantage of using the F1-Score is that it incorporates both precision and recall into a single metric where a high F1-Score means that the model is performing well, even in situations where action classes may be imbalanced. In our algorithm, the F1-Score is computed over actions segments.
- Latency: given a temporal interval between the frame from which the groundtruth action begins and the frame of the predicted label h , and the length of an action H . The latency of this action is defined as [28]:

$$Latency = \frac{h}{H} \quad (9)$$

The latency can be used to evaluate the sensitivity of the method.

4.3 Action Recognition Performance

Table 2 show the average Accuracy and F1-Score of all actions on the OAD dataset. In addition to the previous metrics, in Table 3 we added the average Latency of all actions on the UOW dataset since authors in [28] have only used this metric to evaluate their method. To have a fair evaluation, we used the same training, validation and test sets as defined in [15] and [28] for both datasets. Some works were evaluated by their authors using only accuracy, others using only F1-Score, therefore we put Not Evaluated (N/E) on the metrics that are not provided. On the OAD dataset, we achieved an Accuracy of 0.953 before MV with a very slight improvement with the MV reaching 0.954. As for the UOW dataset, we obtained 0.932 before MV and 0.934 after. We also evaluated our method using the IoU metric defined previously, but since all other works that we compare with did not use it, we will just indicate the results. On the OAD dataset, we achieved an IoU of 0.972 before MV and 0.977 after. On the UOW dataset, we obtained an IoU of 0.934 before MV and 0.958 after. For both datasets, we used an overlapping ratio threshold θ equal to 0.8 to compute the IoU.

We can see that the proposed method achieves the best results on all datasets with all evaluation metrics in comparison with the state-of-the-art methods. Moreover, methods that have used RNN/LSTM such as: JCR-RNN [15], RNN-SW [38], ST-LSTM [21] or MM-MT RNN [19] have revealed poorer performance in comparison to our method. The latter takes into account the temporal dependencies within the skeleton joints across time by integrating information

Table 1: F1-Scores of all actions on the OAD dataset

Actions	JCR-RNN [15]	RNN-SW [38]	RF+ST [2]	MM-MT RNN [19]	SW-GCN [8]	STGCN-SWMV
Drinking	0.574	0.441	0.517	0.538	0.090	0.979
Eating	0.523	0.550	0.645	0.658	0.840	1.000
Writing	0.822	0.859	0.803	0.892	0.920	0.993
Opening cupboard	0.495	0.321	0.555	0.643	0.890	1.000
Washing hands	0.718	0.668	0.860	0.800	0.780	1.000
Opening microwave	0.703	0.665	0.610	0.780	0.780	0.963
Sweeping	0.643	0.590	0.437	0.882	0.930	0.996
Gargling	0.623	0.550	0.722	0.658	0.950	0.984
Trowing trash	0.459	0.674	0.688	0.748	0.880	0.949
Wiping	0.780	0.747	0.977	0.943	0.960	0.973
Overall	0.653	0.600	0.672	0.795	0.900	0.954

Table 2: Average Accuracy and F1-Score on the OAD dataset

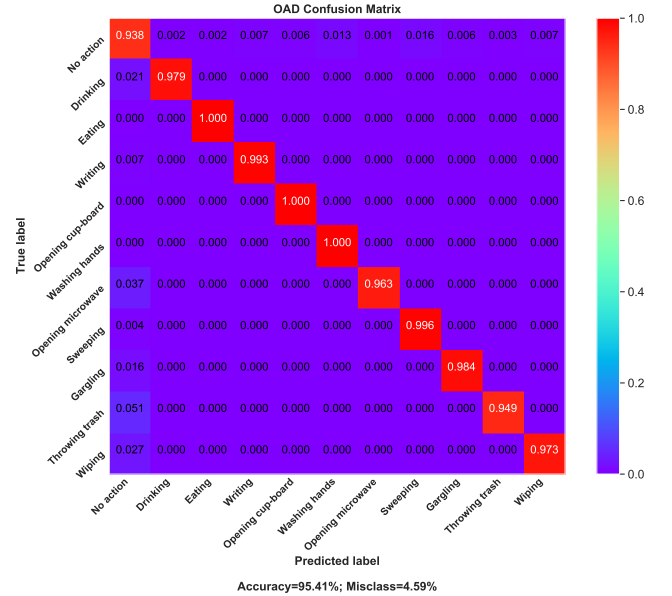
Method	Accuracy	F1-Score
JCR-RNN [15]	N/E	0.653
MD-RNN [18]	N/E	0.628
RNN-SW [38]	N/E	0.600
RF+ST[2]	N/E	0.672
ST-LSTM [21]	0.770	N/E
MM-MT RNN [19]	N/E	0.795
FSNet [20]	0.800	N/E
SSNet [20]	0.820	N/E
MC-LSTM [35]	0.848	N/E
SW-GCN [8]	0.900	0.900
STGCN-SWMV (Ours)	0.954	0.953

Table 3: Average Accuracy, F1-Score and Latency on the UOW dataset

Method	Accuracy	F1-Score	Latency
LE-KSVM [28]	N/E	N/E	0.306
NN [28]	N/E	N/E	0.316
CovaAct [13]	N/E	N/E	0.382
Cov3DJ[28]	N/E	N/E	0.340
SW-CNN [8]	0.680	0.680	N/E
SW-GCN [8]	0.755	0.750	N/E
STGCN-SWMV (Ours)	0.934	0.936	0.047

along both the spatial and the temporal dimension based on the multiple layers of spatial temporal graph convolutions constructed. Afterwards, the temporal and spatial aggregation of the skeleton joints that characterize the performed actions are used to train the STGCN-SWMV module.

Table 1 details the F1-Score of each action class and the overall F1-Score of all actions on the OAD dataset. As it can be seen, our method outperforms all state-of-the-art methods. This shows the effectiveness of the proposed method. Furthermore, we can also see that all actions scores are equally distributed (from 94% to 100%)

**Figure 5: OAD Confusion Matrix with the STGCN-SWMV method**

unlike in [8, 15, 38], where the margin between classes varies from 37% to even 90%.

To have a better grasp of the results, Figure 5 and 6 show the confusion matrices of the F1-Scores on the OAD and UOW datasets respectively.

A few interesting observations can be conducted, in the OAD dataset for example, the "No action" class covers more than 60% of the dataset and sweeps up all the other actions, yet our method still has a good performance on all the other actions. This can explain why all the other actions are mistaken by the "No action" class (See Figure 5). Although the dataset is not equally balanced, has a lot of background noise and actions like "Eating" or "Drinking", for instance, that have similar poses and can be easily mixed up when performing the recognition task, our method has shown its robustness and outperforms other methods in almost all actions.

However, the effectiveness of the proposed method implies a latency of W frames, since a frame must be voted by each SW classifier before taking the decision.

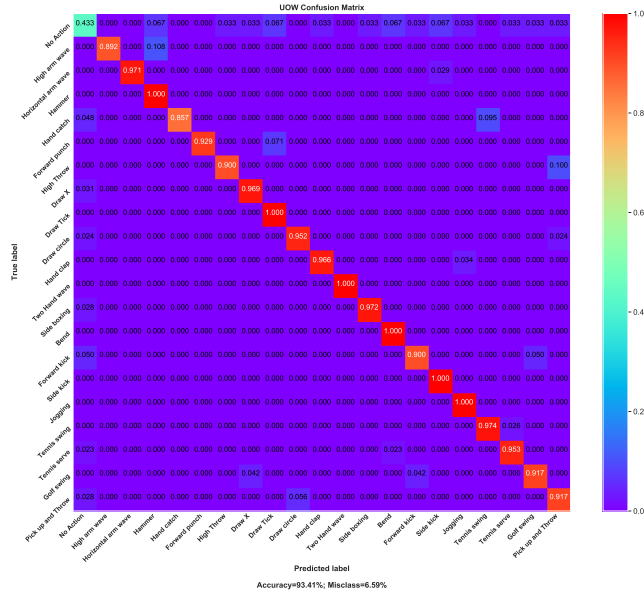


Figure 6: UOW Confusion Matrix with the STGCN-SWMV method

As far as the UOW dataset is concerned, the "No action" class is not well recognized (See Figure 6) and this can be explained by the fact that there is very few samples of this class and with high variability, making it hard for our algorithm to differentiate it from other actions. Moreover, all other actions are very well recognized and our method performed very well on all the them.

Figure 7 and Figure 8 show the timelines comparison of the groundtruth and predicted labels on the OAD and UOW datasets respectively. As it can clearly be seen, predicted labels matches groundtruth ones in almost all frames including the duration as well as the start and end time of each action, which demonstrates the effectiveness of our method and its suitability for Online HAR.

5 CONCLUSION AND FUTURE WORK

In this paper, we propose a novel algorithm for skeleton-based Online HAR using a ST-GCN with a Sliding Window and Majority Voting approach (STGCN-SWMV). We investigated the Online HAR problem for continuous streaming skeletal data which is very challenging and very less developed in the literature in comparison with Segmented HAR. We have shown that our approach allowed greater predictive power and generalization capability overcoming previous methods limitations by automatically learning both spatial and temporal information from data, which makes this method more suitable than others for Online HAR.

Experiments on two challenging large-scale Online skeleton-based datasets demonstrate the effectiveness of our method as it outperforms previous state-of-the-art methods. In future work, to further improve the Online recognition performance, contextual

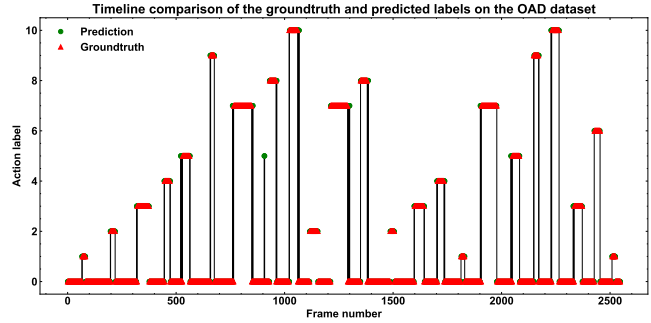


Figure 7: Timeline comparison of the groundtruth and predicted labels on the OAD dataset with the STGCN-SWMV method. Vertical lines denotes frames that are incorrectly predicted.

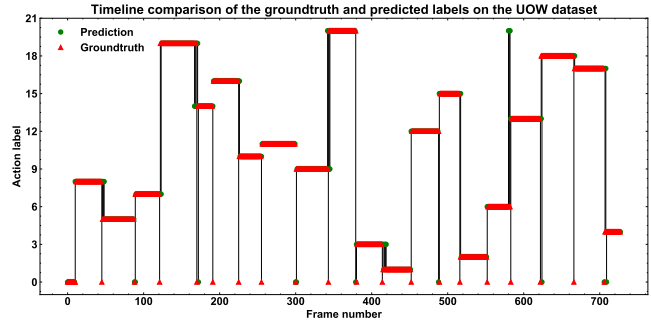


Figure 8: Timeline comparison of the groundtruth and predicted labels on the UOW dataset with the STGCN-SWMV method. Vertical lines denotes frames that are incorrectly predicted.

information such as appearance or human-object interaction could be incorporated, creating more features to learn. Also, using dynamic Sliding Windows can improve results since it can cope with different actions durations.

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